SMALL CLASS 7: The Probability Density Function and Expected Value SOLUTIONS

In this class you will learn about expected value for both discrete and continuous random variables, compute expected value using the PDF, and employ intuitive arguments to check your calculations.

Small class questions

1. **Warm-up and Definitions.**

   (a) A class of 50 students takes an exam. 30 score 70%, and 20 score 50%. What is the class average?

   The average is

   $$\frac{30 \text{ students} \cdot 70\% + 20 \text{ students} \cdot 50\%}{50 \text{ students}} = \frac{\cancel{30 \text{ students}} \cdot 70\% + \cancel{20 \text{ students}} \cdot 50\%}{\cancel{50 \text{ students}}} = \frac{310}{5}\% = 62\%.$$

   We can also write this calculation as

   $$\begin{aligned} &\text{the sum of } \{\text{each score a student can get}\} \times \{\text{the probability of that score}\} \\ &= 70\% \cdot \frac{30}{50} + 50\% \cdot \frac{20}{50} \\ &= 70\% \cdot 0.6 + 50\% \cdot 0.4 = 62\%. \end{aligned}$$

   A student you don't know must write the same test. What score do you "expect" for them?

   If pressed to give a single value, the best answer to this question is probably 62%; that is, the class average is what we call the *expected value*.

   Now, we are using some data to imply the shape of the PDF. Maybe we would want more data before we became confident in our expected value. These details (and how much data you would reasonably need) are something you would dig into in a statistics course. For now, let's suppose we have full info of the PDF and go on to define the expected value.

   Definition: The *Expected value* or *Expectation* is

   - $\mathbb{E}(X) = \sum x \cdot \Pr(X = x)$ (for discrete variables),
   - $\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$ (for continuous variables)

   where: $\Pr(X = x)$ is the probability that the outcome is x (discrete); and $f(x)$ is the PDF (continuous).

   (b) Take a fair 4-sided die. What is the expected value?

   A fair 4-sided die has sides labelled 1, 2, 3, and 4, each of which is equally likely to be rolled. *Equally likely* means that each side has the same probability of getting rolled; with 4 sides, that means each side gets rolled with probability $\frac{1}{4}$ or 0.25. So, the expected value is (notice that we are using the discrete formula, because there are 4 distinct sides):

   $$\mathbb{E}(X) = \sum x \cdot \Pr(X = x) = 1 \cdot 0.25 + 2 \cdot 0.25 + 3 \cdot 0.25 + 4 \cdot 0.25 = 0.25 + 0.5 + 0.75 + 1 = 2.5.$$

   If we roll the die many times, we're expecting an average of 2.5.

2. **Intuitive properties.**

   (a) My friend has a weighted (not fair) 4 sided die with probabilities given by

x	$\Pr(R = x)$
1	0.35
2	0.3
3	0.2
4	0.15

Without doing any calculations, discuss with your group and choose the correct statement:

(A) $\mathbb{E}(X) < 2.5$ (B) $\mathbb{E}(X) = 2.5$ (C) $\mathbb{E}(X) > 2.5$ (D) Not sure...

The probabilities are decreasing, so $\mathbb{E}(X)$ should be less than the expected value of the fair 4 sided die.

(b) Consider the PDF

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}.$$

Without solving any integrals, discuss with your group and choose the correct statement:

(A) $\mathbb{E}(X) < 0$ (B) $0 \le \mathbb{E}(X) < \frac{1}{2}$ (C) $\frac{1}{2} \le \mathbb{E}(X) < 1$ (D) $\mathbb{E}(X) \ge 1$ (E) ??

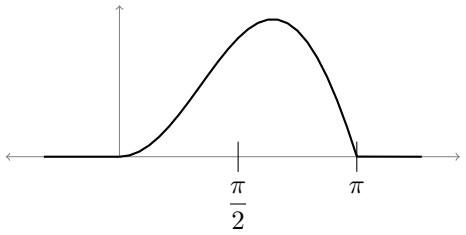
The PDF $f(x)$ is increasing, so $\mathbb{E}(X)$ is after the half-way point.

3. **PDF and expected value: calculations and intuitive arguments.**

(a) A continuous random variable X has PDF

$$f(x) = \begin{cases} kx \sin x & 0 \le x \le \pi \\ 0 & \text{else} \end{cases}$$

for a constant k.



(i) Find k.

Recall from large class: If $f(x)$ is a probability density function then $\int_{-\infty}^{\infty} f(x)\,dx = 1$. Thus,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x)\,dx &= \int_{-\infty}^{0} 0\,dx + \int_{0}^{\pi} kx\sin x\,dx + \int_{\pi}^{\infty} 0\,dx \\ &= k\int_{0}^{\pi} x\sin x\,dx \end{aligned}$$

use IBP:

$u = x,\ dv = \sin x\,dx$

$du = dx,\ v = -\cos x$

$$\begin{aligned} &= k\left[-x\cos x\Big|_0^{\pi} + \int_0^{\pi}\cos x\,dx\right] \\ &= k\left[-\pi\cos\pi + \cancel{0} + \cancel{\sin\pi} - \cancel{\sin 0}\right] \\ &= k\pi, \quad \text{which must be equal to 1. We find } k = \frac{1}{\pi}. \end{aligned}$$

(ii) From the picture, predict whether $\Pr(0 \le X \le \pi/2)$ will be greater or less than $1/2$.

Recall that a probability is equivalent to an area under the PDF curve. The area under the PDF curve as pictured above is not symmetrical; it is "shifted" to the right. That is, the area from 0 to $\pi/2$ is less than the area from $\pi/2$ to π. Since the entire area under the curve must be 1, the area from 0 to $\pi/2$ must be less than $1/2$, so we expect $\Pr(0 \le X \le \pi/2)$ to be less than $1/2$.

(iii) Find $\Pr(0 \le X \le \pi/2)$.

$$\begin{aligned}
\Pr(0 \le X \le \pi/2) &= \int_0^{\pi/2} f(x)\ \mathrm{d}x \\
&= \int_0^{\pi/2} \frac{x}{\pi} \sin x\ \mathrm{d}x = \frac{1}{\pi} \int_0^{\pi/2} x \sin x\ \mathrm{d}x \\
&= \frac{1}{\pi} \left[-x \cos x \Big|_0^{\pi/2} + \int_0^{\pi/2} \cos x\ \mathrm{d}x \right] && \text{using same IBP as before} \\
&= \frac{1}{\pi} \left[-\frac{\pi}{2} \cancel{\cos(\pi/2)} + 0 + \sin(\pi/2) - \cancel{\sin(0)} \right] \\
&= \frac{1}{\pi}.
\end{aligned}$$

Check: Since π is pretty close to 3, we can approximate $1/\pi \approx 1/3$, which is indeed less than $1/2$.

(iv) From the picture, predict whether $\mathbb{E}(X)$ will be greater or less than $\dfrac{\pi}{2}$.

Again, from the picture, we see that more than half of the area under the curve is to the right of $\pi/2$, which means that our expected value should also be greater than $\pi/2$.

(v) Find $\mathbb{E}(X)$.

$$\begin{aligned}
\mathrm{E}(X) &= \int_0^{\pi} x \cdot f(x)\ \mathrm{d}x \\
&= \int_0^{\pi} x \cdot \frac{x}{\pi} \sin x\ \mathrm{d}x = \frac{1}{\pi} \int_0^{\pi} x^2 \sin x\ \mathrm{d}x \\
&\ \ \vdots \qquad \text{using IBP twice! (steps not shown here)} \\
&= \pi - \frac{4}{\pi}.
\end{aligned}$$

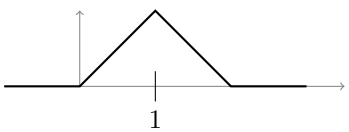
There are many ways we could check this value to make sure it is indeed greater than $\pi/2$, as we predicted, without using a calculator. Here is one:

$$\pi - \frac{4}{\pi} = \left(\frac{\pi}{2} + \frac{\pi}{2}\right) - \frac{4}{\pi} = \frac{\pi}{2} + \left(\frac{\pi}{2} - \frac{4}{\pi}\right) > \frac{\pi}{2} + \left(\frac{3}{2} - \frac{4}{3}\right) = \frac{\pi}{2} + \left(\frac{1}{6}\right) > \pi/2.$$

(b) Consider the PDF

$$f(x) = \begin{cases} x & 0 \le x \le 1 \\ 2 - x & 1 < x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

of a random variable X.



(i) Compute $\mathbb{E}(X)$.

$$\begin{aligned}\mathrm{E}(X) &= \int_0^2 x \cdot f(x)\, \mathrm{d}x \\ &= \int_0^1 x^2\, \mathrm{d}x + \int_1^2 2x - x^2\, \mathrm{d}x \\ &= \frac{1}{3} + 3 - \frac{8}{3} + \frac{1}{3} \\ &= 1.\end{aligned}$$

(ii) How could you have predicted $\mathbb{E}(X)$ without computing it?

The PDF is symmetric around 1 so we predict the expected value should also be 1.

4. **Challenge. Math 101 Final Exam problem, from April 2025.** Let a and b be real numbers and let

$$f(x) = \begin{cases} 0 & \text{if} \quad x < a\,, \\ e^{bx} & \text{if} \quad x \geq a\,. \end{cases}$$

Find (with justification) the values for a and b such that $f(x)$ is a probability density function with expected value equal to 0.

There are two conditions to be satisfied by a and b:

$$\Pr(-\infty < X < \infty) = \int_a^\infty e^{bx}\, \mathrm{d}x = 1 \tag{1}$$

and

$$\mathbb{E}(X) = \int_a^\infty x e^{bx}\, \mathrm{d}x = 0. \tag{2}$$

Note that both are improper integrals.

Looking at the first integral in (1), we note right from the start that we need $b < 0$ because otherwise the improper integral would be divergent.

So, we evaluate the first improper integral in (1) as:

$$\lim_{t\to\infty} \int_a^t e^{bx}\, \mathrm{d}x = \lim_{t\to\infty} \left(\frac{e^{bx}}{b}\right)\Bigg|_a^t = \lim_{t\to\infty} \left(\frac{e^{bt}}{b} - \frac{e^{ba}}{b}\right) = -\frac{e^{ab}}{b}$$

because $e^{bt} \to 0$ as $t \to \infty$ since $b < 0$. So, the condition that $f(x)$ is a probability density function yields that

$$-\frac{e^{ab}}{b} = 1, \text{ i.e. } e^{ab} = -b. \tag{3}$$

Next, we evaluate the expected value integral in (2), using first integration by parts because

$$\int x e^{bx}\, \mathrm{d}x = x \cdot \frac{e^{bx}}{b} - \int \frac{e^{bx}}{b}\, \mathrm{d}x = \frac{x e^{bx}}{b} - \frac{e^{bx}}{b^2} + C.$$

So, we compute the second improper integral (2) from above:

$$\mathbb{E}(X) = \lim_{t\to\infty} \int_a^t x e^{bx}\, \mathrm{d}x = \lim_{t\to\infty} \left(\frac{x e^{bx}}{b} - \frac{e^{bx}}{b^2}\right)\Bigg|_a^t$$

and then using that $b < 0$ and so, both $e^{bt} \to 0$ and $te^{bt} \to 0$, as $t \to \infty$, we get that

$$\mathbb{E}(X) = \lim_{t\to\infty} \left(\frac{te^{bt}}{b} - \frac{e^{bt}}{b^2}\right) - \left(\frac{ae^{ba}}{b} - \frac{e^{ba}}{b^2}\right) = -\frac{ae^{ab}}{b} + \frac{e^{ab}}{b^2}. \tag{4}$$

Next, we note that we know from (3) that $e^{ab} = -b$; substitute this into (4) to get that

$$\mathbb{E}(X) = a - \frac{1}{b}.$$

Then, using that $\mathbb{E}(X) = 0$, we find

$$a - \frac{1}{b} = 0, \text{ i.e. } ab = 1.$$

Now, because $e^{ab} = -b$ (from (3)) and $ab = 1$ (just above), then we have

$$b = -e^{ab} = -e^1 = -e.$$

Finally, because $ab = 1$, we conclude that

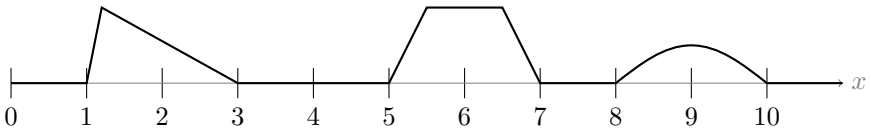
$$a = \frac{1}{b} = \frac{1}{-e} = -\frac{1}{e}.$$

Our final answers are $a = -\frac{1}{e}$ and $b = -e$.

Practice questions See some answers available here; some in the text; or start a conversation on piazza.

The questions below are for practice. They do not contribute to your grade, and it is not expected that you complete them during your small class. However, you are strongly encouraged to work through them as they provide practice for this week's large and small classes.

5. Probability: Math 101 Problem Book, Section 2.1: Q1, Q2, Q3
6. The PDF: Math 101 Problem Book, Section 2.2: Q1-Q7
7. Expected value: Math 101 Problem Book, Section 2.3: Q1-Q13
8. Suppose X is a continuous random variable representing the distance a player reaches on a video game map before their character dies or the game ends. Its PDF $f(x)$ is shown below.



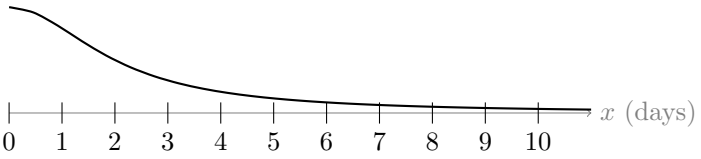
Discuss the following questions.

(a) In which of the intervals below is a player most likely to die?

(A) $1 \leq x \leq 3$
(B) $3 \leq x \leq 5$
(C) $5 \leq x \leq 7$
(D) $8 \leq x \leq 10$

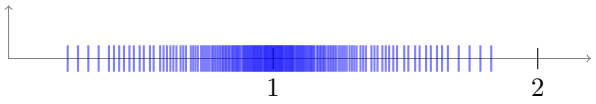
(b) What might be going on in the game from $3 \leq x \leq 5$? Somewhere it's hard/impossible to die, like an animation showing plot development, or a path where you travel but don't encounter dangers.

9. Let X be a continuous random variable representing the length of time after diagnosis with a contagious disease that a person is able to infect others. Its PDF is shown below.

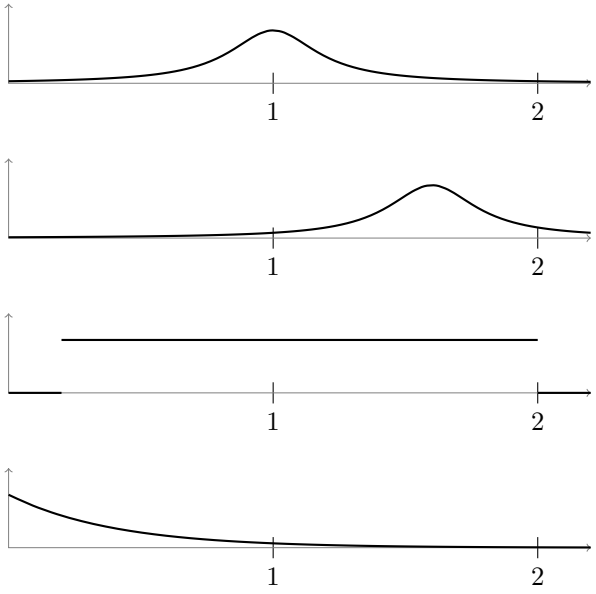


Is a person more likely to be contagious for only 1-2 days after diagnosis, or for 2-3 days after diagnosis?

10. A large number of fish were captured in a river. Each fish was weighed, and its weight recorded as a tick mark on a number line, shown below.



Imagine a continuous random variable X whose value is the weight of a fish captured in the river. Which PDF seems to fit the observed data the best?



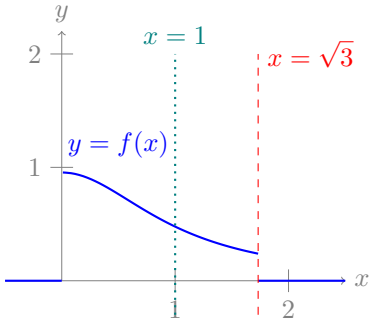
11. Consider the probability density function defined below:

$$f(x) = \begin{cases} 0, & x < 0 \\ \dfrac{3}{\pi}\dfrac{1}{1+x^2}, & 0 \le x \le \sqrt{3} \\ 0, & x > \sqrt{3} \end{cases}.$$

(a) (i) Sketch $f(x)$.

(ii) Use your sketch to determine whether:
$\Pr(0 \le X \le 1) < \Pr(1 \le X \le \sqrt{3})$ OR $\Pr(0 \le X \le 1) > \Pr(1 \le X \le \sqrt{3})$?

Plot of $y = f(x)$ vs x, with $y = f(x)$ in blue, the line $x = 1$ in dotted teal, and $x = \sqrt{3}$ indicated as a red dashed line:



(Your sketch does not have to be exact. The features that your sketch should capture are: your curve on the interval $[0, \sqrt{3}]$ should have a y-intercept less than 1 (because $f(0) = 3/\pi < 1$) and the curve should be decreasing (because the denominator, $1 + x^2$, is increasing). These features are sufficient to answer the question about which probability is larger.)

By inspection, the area under the curve to the *left* of the dotted line $x = 1$ is larger than that to the *right* of $x = 1$, so $\Pr(0 \le X \le 1) > \Pr(1 \le X \le \sqrt{3})$.

(b) Compute the quantities:

(i) $\Pr(0 \le X \le 1)$

$\Pr(0 \le X \le 1) = \displaystyle\int_0^1 f(x)\,\mathrm{d}x = \int_0^1 \frac{3}{\pi}\frac{1}{1+x^2}\,\mathrm{d}x = \frac{3}{\pi}(\arctan(1) - \arctan(0)) = \frac{3}{\pi}\left(\frac{\pi}{4} - 0\right) = \frac{3}{4}.$

(ii) $\Pr(1 \le X \le \sqrt{3})$

$$\begin{aligned}\Pr(1 \le X \le \sqrt{3}) &= \int_1^{\sqrt{3}} f(x)\,\mathrm{d}x = \int_1^{\sqrt{3}} \frac{3}{\pi}\frac{1}{1+x^2}\,\mathrm{d}x \\ &= \frac{3}{\pi}(\arctan(\sqrt{3}) - \arctan(1)) = \frac{3}{\pi}\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{1}{4}.\end{aligned}$$

(This makes sense because $f(x) = 0$ for $x < 0$ and $x > \sqrt{3}$, and we calculated $\Pr(0 \le X \le 1) = \frac{3}{4}$ in (i), so we know that $\Pr(1 \le X \le \sqrt{3}) = 1 - \Pr(0 \le X \le 1) = 1 - \frac{3}{4} = \frac{1}{4}$.)

(iii) $\Pr(-\infty \le X \le \infty)$

By the property in Q2(c), we know that $\Pr(-\infty \le X \le \infty) = \displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$. Let's verify this by direct computation:

$$\begin{aligned}\Pr(-\infty \le X \le \infty) = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x &= \int_{-\infty}^{0} f(x)\,\mathrm{d}x + \int_0^{\sqrt{3}} f(x)\,\mathrm{d}x + \int_{\sqrt{3}}^{\infty} f(x)\,\mathrm{d}x \\ &= \int_{-\infty}^{0} 0\,\mathrm{d}x + \int_0^{\sqrt{3}} \frac{3}{\pi}\frac{1}{1+x^2}\,\mathrm{d}x + \int_{\sqrt{3}}^{\infty} 0\,\mathrm{d}x \\ &= \int_0^{\sqrt{3}} \frac{3}{\pi}\frac{1}{1+x^2}\,\mathrm{d}x \\ &= \frac{3}{\pi}(\arctan\sqrt{3} - \arctan 0) \\ &= \frac{3}{\pi}\frac{\pi}{3} = 1.\end{aligned}$$

Alternatively, since we already calculated $\Pr(0 \le X \le 1)$ and $\Pr(1 \le X \le \sqrt{3})$, we could simply write:

$$\begin{aligned}\Pr(-\infty \le X \le \infty) = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x &= \int_{-\infty}^{0} f(x)\,\mathrm{d}x + \int_0^{1} f(x)\,\mathrm{d}x + \int_1^{\sqrt{3}} f(x)\,\mathrm{d}x + \int_{\sqrt{3}}^{\infty} f(x)\,\mathrm{d}x \\ &= \int_{-\infty}^{0} 0\,\mathrm{d}x + \Pr(0 \le X \le 1) + \Pr(1 \le X \le \sqrt{3}) + \int_{\sqrt{3}}^{\infty} 0\,\mathrm{d}x \\ &= 0 + \frac{3}{4} + \frac{1}{4} + 0 = 1.\end{aligned}$$

12. Suppose the function $f(x) = \dfrac{A}{x^2+4}$ is a probability density function (PDF). What is A?

Hint: For $f(x)$ to be a PDF, we require $\displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$.

Answer: $A = \dfrac{2}{\pi}$.